ACHELION CAPITAL MANAGEMENT, LLC

# ARMS End-of-Day Snapshot

Specification — Addendum to Infrastructure Specification v1.0

For: Josh Paul — Lead Developer

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**Classification: CONFIDENTIAL — GP and Development Use Only**

**The morning monitor is the regime and thesis instrument — read at 0800 CT before the market opens. The End-of-Day Snapshot is the execution and positioning instrument — read at 1455 CT before the close. Two different purposes. Two different templates. Same pipeline.**

*"Read clean before the open. Confirm clean before the close."*

# 0. Purpose and Design Philosophy

The daily ARMS monitor is generated at 0530 CT from the overnight data snapshot and delivered at 0600 CT. The PM reads it at 0800 CT — 90 minutes before market open. At that point the monitor reflects pre-market conditions and everything the system has assessed overnight. It is the right instrument for its purpose: regime orientation, thesis status, module health, and the decision queue entering the day.

But markets move during the session. Earnings calls happen at 0900. Regulatory announcements drop at 1000. CDM signals fire throughout the day. Tier 1 veto windows open and close. LAEP executes orders. By 1455 CT — five minutes before the close — the portfolio may be in a materially different state than the morning monitor described.

The End-of-Day Snapshot addresses this. It is not a second full monitor. It is a focused five-field closing status check that tells the PM and GPs exactly what changed during the session, what executed, what is still pending, and what the regime looks like entering the overnight period. It reads in under two minutes. It requires no action in most sessions. When it does require action it surfaces a single specific item — not a general alert.

**The morning monitor sets the day. The End-of-Day Snapshot closes it. Together they constitute a complete daily information architecture with no gap between the open and the close.**

# 1. Schedule and Delivery

| | |
|---|---|
| **Generation time** | 1450 CT — 10 minutes before close. Captures full session activity including final hour moves. |
| **Delivery time** | 1455 CT — 5 minutes before close. Both PM and GPs receive simultaneously. |
| **Recipients** | MJ Pence (PM) · Brandon [Last Name] (GP) · additional GPs added via config as they join |
| **Format** | Same dual-format as morning monitor: web dashboard refresh + PDF email delivery via SendGrid or Gmail SMTP |
| **Scheduler entry** | Prefect flow: eod_snapshot_flow() scheduled at 1450 CT Monday through Friday. Skips US market holidays automatically via trading calendar check. |
| **File** | src/reporting/eod_snapshot.py — new file. Separate template from daily_monitor.py. Shares the same data pipeline and module outputs. |
| **Session log** | Every EOD Snapshot generation logged with timestamp, regime score at generation, and delta from morning monitor score. |

# 2. The Five Fields — Complete Specification

The End-of-Day Snapshot contains exactly five fields. No more. The discipline of five fields is intentional — if a sixth field is ever proposed, the correct response is to ask whether the morning monitor should be updated instead. The EOD Snapshot is a closing confirmation instrument, not a second full monitor.

## Field 1 — Regime Score Delta

**Has the regime score moved materially since 0800 CT? If yes, what drove it?**

| | |
|---|---|
| **Content** | Current regime label + current score · Morning score (from 0600 CT monitor) · Delta (current minus morning) · Direction arrow · One-sentence narrative if delta exceeds threshold |
| **Materiality threshold** | Delta of ±0.03 or greater triggers a narrative. Below ±0.03 displays as 'Stable — no material movement during session.' |

| | |
|---|---|
| **Color coding** | Green if score improved (moved toward RISK_ON) · Red if score worsened (moved toward CRASH) · Grey if stable |
| **CRASH boundary alert** | If current score is within 0.05 of the CRASH boundary (0.80), this field displays a RED border regardless of delta direction — proximity to boundary is always surfaced |
| **Data source** | ARAS module output at 1450 CT generation time |
| **Example — stable** | DEFENSIVE · Score: 0.76 · Morning: 0.76 · Delta: 0.00 → Stable — no material movement during session. |
| **Example — worsening** | DEFENSIVE · Score: 0.79 · Morning: 0.76 · Delta: +0.03 ↑ · Iran formally enacted Hormuz toll legislation at 1115 CT. CRASH boundary now 1 point away. |
| **Example — improving** | DEFENSIVE · Score: 0.71 · Morning: 0.76 · Delta: −0.05 ↓ · Peace talks confirmed via Pakistani mediators. Brent fell 4%. RPE: NEUTRAL probability rising. |

## Field 2 — Pending Tier 1 Windows

**Are any Tier 1 veto windows still open that expire at or after market close?**

| | |
|---|---|
| **Content** | List of all open Tier 1 confirmation queue items with their action type, time queued, time remaining in veto window, and current status |
| **If none pending** | Single green line: 'No Tier 1 actions pending. Decision queue clear.' |
| **If items pending** | Each item listed with: action type · position or module · time queued · time remaining · response options reminder |
| **Urgency flag** | Any item with less than 60 minutes remaining in its veto window displays in RED with a countdown. This is the only case where the EOD Snapshot may require immediate PM action. |
| **Silence reminder** | Each pending item includes: 'No response by [expiry time] CT = system executes at MICS-implied sizing.' |
| **Data source** | src/execution/confirmation_queue.py — open items query at 1450 CT |
| **Example** | PERM — VRT covered call evaluation · Queued: 1023 CT · Window expires: 1423 CT · STATUS: EXPIRED — system executed at 1423 CT. No action required. |
| **Example — urgent** | TDC thesis review — ALAB · Queued: 1318 CT · Window expires: 1718 CT · 2h 23m remaining · Response options: [No new info — execute] [New info — hold] [Veto with rationale] |

## Field 3 — Intraday CDM and TDC Signals

**Did any new cross-sector signals or thesis integrity events fire during the session that were not in the morning monitor?**

| | |
|---|---|
| **Content** | Any CDM named entity alerts or TDC thesis reviews triggered between 0600 CT monitor generation and 1450 CT EOD generation |
| **If none** | Single green line: 'No new CDM or TDC signals during session.' |
| **If signals present** | Each signal listed with: signal type · named entity · affected positions · severity · TIS score if TDC ran · recommended action |
| **Severity threshold** | CRITICAL and HIGH signals always displayed. MEDIUM signals displayed only if they affect a position currently flagged WATCH or above. |
| **Data source** | src/engine/cdm.py and src/engine/tdc.py session logs — events between 0600 and 1450 CT |
| **Example** | CDM HIGH: Microsoft announced 15% reduction in Azure AI capex guidance in earnings call at 0932 CT. Affected positions: NVDA, MU, ALAB, ANET, MRVL. TDC thesis reviews queued and completed for all 5 positions. Results: NVDA INTACT (8.4) · MU WATCH (6.8) · ALAB IMPAIRED (4.9) · ANET INTACT (7.9) · MRVL WATCH (6.5). ALAB IMPAIRED: PM review queued — see Field 2. |

## Field 4 — Session Execution Summary

**What did ARMS execute during the session? What is the current portfolio composition versus the target?**

| | |
|---|---|
| **Content** | Summary of all order executions during the session · Any execution failures · Current portfolio weights versus Architecture AB targets · Any weight drift above 0.5pp from target |
| **If no executions** | Single line: 'No orders executed during session. Portfolio weights within tolerance.' |
| **Execution summary format** | [Time CT] [Module] [Action] [Ticker] [Size] [Price] [Status: FILLED / PARTIAL / FAILED] |
| **Weight drift** | Any position where current weight deviates more than 0.5pp from target weight is flagged. Not an action item — an information item. The system will correct at next regime cycle. |
| **Execution failures** | Any FAILED execution is displayed in RED with the error reason. This is the one execution field that may require PM attention — a failed order means a target allocation is not yet achieved. |

| | |
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| **LAEP mode** | Current LAEP execution mode displayed: NORMAL / ELEVATED / CRISIS. If mode changed during session, shows: 'LAEP mode changed: ELEVATED → NORMAL at 1134 CT (VIX crossed below 25).' |
| **Data source** | src/execution/broker_api.py order log · src/engine/master_engine.py current weights · LAEP module output |
| **Example — clean session** | No orders executed during session. Portfolio weights within tolerance. LAEP: ELEVATED throughout session. No execution failures. |
| **Example — active session** | 1023 CT · PERM · SELL CALL · VRT · 100 contracts · $4.20 premium · FILLED. 1245 CT · ARAS · REGIME RESIZE · QQQ PUTS · Notional +$12,400 · FILLED. Portfolio weights: all within 0.3pp of target. |

## Field 5 — Overnight Risk Assessment

**What is the most important thing to watch between now and the 0800 CT morning review?**

| | |
|---|---|
| **Content** | A single AI-generated two to three sentence overnight watch statement. Not a list. Not a general market summary. One specific, actionable statement about the highest-priority risk or opportunity entering the overnight period. |
| **Generation** | Claude API call via claude_wrapper.py. Input: current regime score + delta + any open CDM/TDC signals + PTRH status + any pending Tier 1 items. Output: two to three sentence watch statement. |
| **Scope constraint** | The watch statement must reference a specific named signal, entity, or threshold. Generic statements ('markets remain volatile') are rejected by a validation step that checks for named specificity. If the Claude API returns a generic statement, the system retries once with a stricter prompt before falling back to a rule-based statement. |
| **Prompt structure** | System: You are the ARMS risk assessment engine. Generate a two to three sentence overnight watch statement for the PM and GPs. Be specific — name the entity, signal, or threshold that matters most. Do not summarize the session. Identify the single most important thing to watch between now and 0800 CT tomorrow. Current state: [regime + delta + signals + pending items]. |
| **Data source** | Claude API · all module outputs at 1450 CT generation |
| **Example — war scenario** | Watch: Iran's response to Trump's 1400 CT ultimatum is expected via state media overnight. If Iran formally announces Hormuz toll rates above $500,000 per vessel, the regime score will likely cross 0.80 at the CRASH boundary. Confirm PTRH dual-risk coverage before tomorrow's open. |

| | |
|---|---|
| **Example — stable session** | Watch: Regime score held stable at 0.71 through the full session. The primary overnight catalyst is NVDA's earnings call scheduled for 1700 CT — capex commitment language in the hyperscaler discussion will update MU and ALAB TIS scores before tomorrow's morning monitor. No action required tonight. |
| **Example — earnings watch** | Watch: Microsoft reports earnings at 1600 CT today. ELVT will process the transcript overnight and update language velocity scores for MU, NVDA, ALAB, ANET, and MRVL. Any material capex language shift will be reflected in tomorrow's morning monitor. JPVI Microsoft Azure infrastructure hiring trend will be cross-referenced automatically. |

# 3. Visual Layout Specification

The EOD Snapshot uses the same CSS design system and color variables as the morning monitor. The layout is a single vertical column — five cards stacked top to bottom, each self-contained. No tables, no multi-column layout. The PM reads top to bottom in under two minutes.

| Position | Field | Card style | Height | Normal-session read time |
|---|---|---|---|---|
| 1 | Regime Score Delta | Full-width card. Score bar. Delta indicator. Color coded green/red/grey. | Compact — 80px | 5 seconds |
| 2 | Pending Tier 1 Windows | Full-width card. Green if clear. Amber/red if items pending with countdown. | Compact if clear · Expanded if items present | 5 seconds if clear · 20 seconds if items present |
| 3 | Intraday CDM/TDC Signals | Full-width card. Green if clean. Amber/red with signal details if signals fired. | Compact if clean · Expanded if signals present | 5 seconds if clean · 30 seconds if signals present |
| 4 | Session Execution Summary | Full-width card. Green if no executions or clean executions. Red border if any FAILED orders. | Medium — execution list if orders present | 10 seconds |
| 5 | Overnight Risk Assessment | Full-width card. Gold left border. Two to three sentence watch statement. Always present. | Fixed — 100px | 10 seconds |

Header: identical to morning monitor but with session date, generation time (1450 CT), and a distinct badge: 'END-OF-DAY SNAPSHOT — 1455 CT'. Footer: identical to morning monitor.

CRITICAL DESIGN RULE: The EOD Snapshot must never expand to more than one printed page when rendered as PDF. If any field generates content that would push the layout beyond one page, the content is truncated with a 'Full detail in session log' reference. The one-page constraint is enforced at the template level, not at the data level. The PM reads a one-page closing confirmation, not a second full monitor.

# 4. Implementation Specification

## 4.1 New File

```
# src/reporting/eod_snapshot.py
# Generates the ARMS End-of-Day Snapshot
# Called by Prefect scheduler at 1450 CT M-F
# Outputs: HTML (web dashboard refresh) + PDF (email delivery)

def generate_eod_snapshot() -> EODSnapshotOutput:

    # Field 1 — Regime Score Delta
    morning_score = db.get_morning_monitor_score(today)
    current_score = aras.get_current_composite_score()
    delta         = current_score - morning_score
    narrative     = generate_delta_narrative(delta) if abs(delta) >= 0.03 else None

    # Field 2 — Pending Tier 1 Windows
    pending_items = confirmation_queue.get_open_items()

    # Field 3 — Intraday CDM/TDC Signals
    intraday_signals = db.get_cdm_tdc_events(
        since=morning_monitor_generation_time,
        severity=['CRITICAL', 'HIGH', 'MEDIUM']
    )

    # Field 4 — Session Execution Summary
    session_orders = broker_api.get_session_orders(today)
    weight_drift   = master_engine.get_weight_drift()
    laep_status    = laep.get_session_summary()

    # Field 5 — Overnight Risk Assessment
    watch_statement = claude_wrapper.call(
        task_type='overnight_watch',
        context={
            'regime_score': current_score,
            'score_delta': delta,
            'pending_items': pending_items,
            'intraday_signals': intraday_signals,
            'ptrh_status': tail_hedge.get_status()
        }
    )
```

```
    # Validate overnight watch statement for named specificity
    if not contains_named_entity(watch_statement):
        watch_statement = claude_wrapper.call(task_type='overnight_watch_strict', ...)

    # Assemble and render
    snapshot = EODSnapshotOutput(
        regime_delta=RegimeDelta(morning_score, current_score, delta, narrative),
        pending_tier1=pending_items,
        intraday_signals=intraday_signals,
        execution_summary=ExecutionSummary(session_orders, weight_drift, laep_status),
        overnight_watch=watch_statement,
        generated_at=datetime.now(CT).isoformat()
    )

    html = render_eod_template(snapshot)        # one-page constraint enforced here
    pdf  = weasyprint.HTML(string=html).write_pdf()

    deliver_eod_snapshot(html, pdf, RECIPIENTS)
    db.log_eod_snapshot(snapshot)
    return snapshot
```

## 4.2 Scheduler Entry

```
# Add to Prefect scheduler configuration
@flow(name='eod_snapshot_daily')
def eod_snapshot_flow():
    if not is_market_day(today):  # skip weekends + US holidays
        return
    generate_eod_snapshot()

# Schedule: 1450 CT Monday through Friday
# Cron: '50 14 * * 1-5' (UTC: '50 20 * * 1-5')
```

## 4.3 Delivery Configuration

```
# Add EOD recipients alongside morning monitor recipients
EOD_RECIPIENTS = [
    'mj@achelion.com',       # PM — MJ Pence
    'brandon@achelion.com'   # GP — Brandon [Last Name]
    # Add Crystal and James when they join as active GPs
]

EOD_EMAIL_SUBJECT = 'ARMS End-of-Day Snapshot — {date} — {regime} {score}'
# Example: 'ARMS End-of-Day Snapshot — March 26, 2026 — DEFENSIVE 0.76'
```

### 4.4 Updated Files

| | |
|---|---|
| **src/reporting/eod_snapshot.py** | NEW — full EOD Snapshot generation logic as specified above |
| **src/reporting/daily_monitor.py** | UNCHANGED — morning monitor is a separate template and schedule. No changes required. |
| **src/config/delivery_config.py** | UPDATED — add EOD_RECIPIENTS list and EOD_EMAIL_SUBJECT template |
| **src/scheduling/prefect_flows.py** | UPDATED — add eod_snapshot_flow() at 1450 CT cron schedule |
| **src/intelligence/claude_wrapper.py** | UPDATED — add 'overnight_watch' and 'overnight_watch_strict' task types with associated prompt templates |

## 5. The Complete Daily Information Architecture

**After both the morning monitor and the EOD Snapshot are live, this is the PM's complete daily information schedule.**

| Time CT | Event | Instrument | What it tells you |
|---|---|---|---|
| 0600 | Daily monitor delivered | Full morning monitor | Complete regime assessment. All positions. All module status. Decision queue. Overnight developments processed. Pre-market picture. |
| 0800 | PM reads morning monitor | Full morning monitor | Regime orientation. Thesis status. Module health. Decision queue for the day. 90 minutes before open — time to think without execution pressure. |
| 0930 | Market opens | System autonomous | ARMS is running. LAEP in ELEVATED or NORMAL mode. CDM and TDC monitoring continuously. No PM involvement unless a Category A question surfaces. |
| All session | Continuous monitoring | System autonomous | CDM/TDC signals propagate in real time. PTRH rolls execute if triggered. PERM executes if triggered. Regime re-scores every 5 minutes. PM not involved. |
| 1450 | EOD Snapshot generated | 5-field closing check | Regime delta. Pending Tier 1 windows. Intraday signals. Execution summary. Overnight watch statement. |

| | | | |
|---|---|---|---|
| 1455 | PM reads EOD Snapshot | 5-field closing check | What changed during the session. What executed. What is still pending. What to watch overnight. Two minutes to read. |
| 1600 | Market closes | System autonomous | Session closes. ARMS continues monitoring overnight data. Regime scoring continues on overseas markets and macro feeds. |
| 2000 CT | Knowledge base nightly ingest | Background process | All session data embedded into Chroma DB. Tomorrow's morning monitor will have full context. |
| 0530 | Morning monitor generated | Full morning monitor | Next day's cycle begins. Overnight developments assessed. Regime scored. All modules run. Monitor assembled. |

Total PM time required on a normal day: approximately 12 minutes. Eight minutes reading the morning monitor at 0800. Two minutes reading the EOD Snapshot at 1455. Two minutes for any Category A response if the system surfaced one. On most days, the PM does nothing between 0802 and 1453. The system runs the portfolio. The PM runs Achelion.

**Morning monitor: regime and thesis. EOD Snapshot: execution and confirmation. Together: a complete daily information architecture with zero gaps and zero redundancy.**

*"Read clean before the open. Confirm clean before the close."*

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